

Thiết Kế Cơ Sở Dữ Liệu

Đề Tài: LivingDocs

Nhóm 1

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Preface

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part I.

First half of the book

chapter 1.

Introduction

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1.1. Research question

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¹Excepteur sint, sunt in culpa qui officia deserunt mollit anim id est laborum. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.2. Answer to the question

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.² Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

1.3. Positioning in the literature

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1.4. A section

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²Sunt in culpa qui officia deserunt mollit anim id est laborum. Excepteur sint occaecat cupidatat non proident. Sunt in culpa qui officia deserunt mollit anim id est laborum. Excepteur sint occaecat cupidatat non proident. Sunt in culpa qui officia deserunt mollit anim id est laborum. Excepteur sint occaecat cupidatat non proident.

³Obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio, cumque nihil impedit, quo minus id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

1.4.1. A subsection

Maecenas nec risus nec nibh hendrerit ultricies. Nulla facilisi. Maecenas in magna eget felis dictum tempus. Donec vehicula, dolor nec dictum aliquam, arcu libero commodo metus, nec varius neque magna at nulla. Sed nec arcu sollicitudin, scelerisque tortor nec, viverra arcu. Cras suscipit mi vitae erat varius, nec ullamcorper lectus tristique. Maecenas aliquet lorem vitae velit ullamcorper, eget pellentesque velit rhoncus. Sed at risus at arcu hendrerit facilisis. Sed a massa hendrerit, rutrum orci ac, condimentum enim. Donec eget vestibulum eros. In hac habitasse platea dictumst.

1.4.2. Yet another subsection

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1.5. A final introduction section

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et sapien lacinia eleifend. Curabitur sit amet velit eu metus suscipit tempus. Maecenas in libero felis. Sed quis eros sapien. Suspendisse potenti. Curabitur porta est sed enim efficitur, a tempor justo fermentum. Vivamus pulvinar enim vitae semper rutrum. Aenean non mauris sed lectus vestibulum feugiat eget et ipsum. Donec semper vitae enim at ultricies. Sed eu arcu justo.

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chapter 2.

Sections and content

part II.

Thiết kế & Cài đặt

chapter 3.

Thuật toán maping ER

3.1. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

3.1.1. Cú pháp Rút gọn (Logical Schema Syntax)

- users (user_id UUID PK, email VARCHAR(255), password_hash VARCHAR(255), first_name VARCHAR(100), last_name VARCHAR(100), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)[cite: 1]
- workspaces (workspace_id UUID PK, name VARCHAR(150), description TEXT, status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)[cite: 1]
- roles (role_id UUID PK, role_name VARCHAR(50), description TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)[cite: 1]
- permissions (permission_id UUID PK, permission_code VARCHAR(50), description TEXT, created_at TIMESTAMP)[cite: 1]
- github_connections (connection_id UUID PK, user_id UUID FK, github_user_id VARCHAR(50), github_username VARCHAR(100), access_token VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)[cite: 1]

- `user_notification_preferences` (preference_id UUID PK, user_id UUID FK, channel VARCHAR(20))[cite: 1]
- `github_connection_scopes` (scope_id UUID PK, connection_id UUID FK, scope_value VARCHAR(50))[cite: 1]
- `workspace_members` (membership_id UUID PK, user_id UUID FK, workspace_id UUID FK, role_id UUID FK, joined_at TIMESTAMP, status VARCHAR(20))[cite: 1]
- `role_permissions` (role_permission_id UUID PK, role_id UUID FK, permission_id UUID FK, granted_at TIMESTAMP)[cite: 1]

3.1.2. Mã DDL SQL Hoàn Chỉnh (schema_member_1.sql)

```

1
2 CREATE TABLE users (
3     user_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
4     email            VARCHAR(255)  NOT NULL,
5     password_hash    VARCHAR(255)  NULL,
6     first_name       VARCHAR(100)   NULL,
7     last_name        VARCHAR(100)   NULL,
8     status           VARCHAR(20)    NOT NULL,
9     created_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
10    updated_at        TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
11    deleted_at        TIMESTAMP      NULL,
12    CONSTRAINT pk_users PRIMARY KEY (user_id),
13    CONSTRAINT uq_users_email UNIQUE (email)
14 );
15
16
17 CREATE TABLE workspaces (
18     workspace_id     UUID          NOT NULL DEFAULT gen_random_uuid(),
19     name             VARCHAR(150)   NOT NULL,
20     description       TEXT          NULL,
21     status           VARCHAR(20)    NOT NULL,
22     created_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
23     updated_at        TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
24     deleted_at        TIMESTAMP      NULL,
25     CONSTRAINT pk_workspaces PRIMARY KEY (workspace_id)
26 );
27
28
29 CREATE TABLE roles (
30     role_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
31     role_name        VARCHAR(50)    NOT NULL,
32     description       TEXT          NULL,

```

```

33     created_at      TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
34     updated_at      TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
35     CONSTRAINT pk_roles PRIMARY KEY (role_id)
36 );
37
38
39 CREATE TABLE permissions (
40     permission_id    UUID           NOT NULL DEFAULT gen_random_uuid(),
41     permission_code   VARCHAR(50)    NOT NULL,
42     description       TEXT           NULL,
43     created_at        TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
44     CONSTRAINT pk_permissions PRIMARY KEY (permission_id),
45     CONSTRAINT uq_permissions_code UNIQUE (permission_code)
46 );
47
48
49 CREATE TABLE github_connections (
50     connection_id     UUID           NOT NULL DEFAULT gen_random_uuid(),
51     user_id           UUID           NOT NULL,
52     github_user_id    VARCHAR(50)    NOT NULL,
53     github_username    VARCHAR(100)   NOT NULL,
54     access_token       VARCHAR(500)   NOT NULL,
55     created_at         TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
56     updated_at         TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
57     CONSTRAINT pk_github_connections PRIMARY KEY (connection_id),
58     CONSTRAINT uq_github_connections_user UNIQUE (user_id),
59     CONSTRAINT fk_github_connections_user
60         FOREIGN KEY (user_id) REFERENCES users (user_id)
61         ON UPDATE CASCADE
62         ON DELETE CASCADE
63 );
64
65
66 CREATE TABLE user_notification_preferences (
67     preference_id      UUID           NOT NULL DEFAULT gen_random_uuid(),
68     user_id            UUID           NOT NULL,
69     channel            VARCHAR(20)    NOT NULL,
70     CONSTRAINT pk_user_notification_preferences PRIMARY KEY
71     (preference_id),
72     CONSTRAINT fk_user_notification_pref_user
73         FOREIGN KEY (user_id) REFERENCES users (user_id)
74         ON UPDATE CASCADE
75         ON DELETE CASCADE
76 );
77
78 CREATE TABLE github_connection_scopes (

```

```

79     scope_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
80     connection_id     UUID          NOT NULL,
81     scope_value        VARCHAR(50)   NOT NULL,
82     CONSTRAINT pk_github_connection_scopes PRIMARY KEY (scope_id),
83     CONSTRAINT fk_github_connection_scopes_conn
84         FOREIGN KEY (connection_id) REFERENCES github_connections
85         (connection_id)
86         ON UPDATE CASCADE
87         ON DELETE CASCADE
88 );
89 CREATE TABLE workspace_members (
90     membership_id     UUID          NOT NULL DEFAULT gen_random_uuid(),
91     user_id           UUID          NOT NULL,
92     workspace_id      UUID          NOT NULL,
93     role_id           UUID          NOT NULL,
94     joined_at         TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
95     status            VARCHAR(20)   NOT NULL,
96     CONSTRAINT pk_workspace_members PRIMARY KEY (membership_id),
97
98     CONSTRAINT uq_workspace_members_user_workspace UNIQUE (user_id,
99     workspace_id),
100    CONSTRAINT fk_workspace_members_user
101        FOREIGN KEY (user_id) REFERENCES users (user_id)
102        ON UPDATE CASCADE
103        ON DELETE CASCADE,
104    CONSTRAINT fk_workspace_members_workspace
105        FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
106        ON UPDATE CASCADE
107        ON DELETE CASCADE,
108    CONSTRAINT fk_workspace_members_role
109        FOREIGN KEY (role_id) REFERENCES roles (role_id)
110        ON UPDATE CASCADE
111        ON DELETE RESTRICT
112 );
113
114 CREATE TABLE role_permissions (
115     role_permission_id UUID          NOT NULL DEFAULT gen_random_uuid(),
116     role_id            UUID          NOT NULL,
117     permission_id      UUID          NOT NULL,
118     granted_at         TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
119     CONSTRAINT pk_role_permissions PRIMARY KEY (role_permission_id),
120     CONSTRAINT uq_role_permissions_pair UNIQUE (role_id, permission_id),
121     CONSTRAINT fk_role_permissions_role
122         FOREIGN KEY (role_id) REFERENCES roles (role_id)
123         ON UPDATE CASCADE

```

```

124         ON DELETE CASCADE,
125     CONSTRAINT fk_role_permissions_permission
126         FOREIGN KEY (permission_id) REFERENCES permissions (permission_id)
127         ON UPDATE CASCADE
128         ON DELETE CASCADE
129 );
130
131
132 CREATE INDEX idx_github_connections_user_id ON github_connections(user_id);
133 CREATE INDEX idx_workspace_members_workspace_id ON
134     workspace_members(workspace_id);
135 CREATE INDEX idx_workspace_members_user_id ON workspace_members(user_id);
136 CREATE INDEX idx_role_permissions_role_id ON role_permissions(role_id);
137
138 CREATE OR REPLACE FUNCTION set_updated_at()
139 RETURNS TRIGGER AS $$
140 BEGIN
141     NEW.updated_at := CURRENT_TIMESTAMP;
142     RETURN NEW;
143 END;
144
145
146 CREATE TRIGGER trg_users_updated_at
147     BEFORE UPDATE ON users
148     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
149
150 CREATE TRIGGER trg_workspaces_updated_at
151     BEFORE UPDATE ON workspaces
152     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
153
154 CREATE TRIGGER trg_roles_updated_at
155     BEFORE UPDATE ON roles
156     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
157
158 CREATE TRIGGER trg_github_connections_updated_at
159     BEFORE UPDATE ON github_connections
160     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
161
162 GO

```

1.2. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

1.2.1. Cú pháp Rút gọn (Logical Schema Syntax)

- `users` (`user_id` UUID PK, `email` VARCHAR(255), `password_hash` VARCHAR(255), `first_name` VARCHAR(100), `last_name` VARCHAR(100), `status` VARCHAR(20), `created_at` TIMESTAMP, `updated_at` TIMESTAMP, `deleted_at` TIMESTAMP)[cite: 1]
- `workspaces` (`workspace_id` UUID PK, `name` VARCHAR(150), `description` TEXT, `status` VARCHAR(20), `created_at` TIMESTAMP, `updated_at` TIMESTAMP, `deleted_at` TIMESTAMP)[cite: 1]
- `roles` (`role_id` UUID PK, `role_name` VARCHAR(50), `description` TEXT, `created_at` TIMESTAMP, `updated_at` TIMESTAMP)[cite: 1]
- `permissions` (`permission_id` UUID PK, `permission_code` VARCHAR(50), `description` TEXT, `created_at` TIMESTAMP)[cite: 1]
- `github_connections` (`connection_id` UUID PK, `user_id` UUID FK, `github_user_id` VARCHAR(50), `github_username` VARCHAR(100), `access_token` VARCHAR(500), `created_at` TIMESTAMP, `updated_at` TIMESTAMP)[cite: 1]
- `user_notification_preferences` (`preference_id` UUID PK, `user_id` UUID FK, `channel` VARCHAR(20))[cite: 1]
- `github_connection_scopes` (`scope_id` UUID PK, `connection_id` UUID FK, `scope_value` VARCHAR(50))[cite: 1]
- `workspace_members` (`membership_id` UUID PK, `user_id` UUID FK, `workspace_id` UUID FK, `role_id` UUID FK, `joined_at` TIMESTAMP, `status` VARCHAR(20))[cite: 1]
- `role_permissions` (`role_permission_id` UUID PK, `role_id` UUID FK, `permission_id` UUID FK, `granted_at` TIMESTAMP)[cite: 1]

1.2.2. Mã DDL SQL Hoàn Chỉnh (schema_member_1.sql)

```
1
2 CREATE TABLE users (
3     user_id          UUID          NOT NULL DEFAULT gen_random_uuid(),
4     email            VARCHAR(255)  NOT NULL,
5     password_hash    VARCHAR(255)  NULL,
6     first_name       VARCHAR(100)   NULL,
7     last_name        VARCHAR(100)   NULL,
8     status           VARCHAR(20)    NOT NULL,
9     created_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
10    updated_at       TIMESTAMP      NOT NULL DEFAULT CURRENT_TIMESTAMP,
11    deleted_at       TIMESTAMP      NULL,
```

```

12     CONSTRAINT pk_users PRIMARY KEY (user_id),
13     CONSTRAINT uq_users_email UNIQUE (email)
14 );
15
16
17 CREATE TABLE workspaces (
18     workspace_id    UUID                NOT NULL DEFAULT gen_random_uuid(),
19     name             VARCHAR(150)        NOT NULL,
20     description      TEXT                NULL,
21     status           VARCHAR(20)         NOT NULL,
22     created_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
23     updated_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
24     deleted_at       TIMESTAMP            NULL,
25     CONSTRAINT pk_workspaces PRIMARY KEY (workspace_id)
26 );
27
28
29 CREATE TABLE roles (
30     role_id          UUID                NOT NULL DEFAULT gen_random_uuid(),
31     role_name        VARCHAR(50)         NOT NULL,
32     description      TEXT                NULL,
33     created_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
34     updated_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
35     CONSTRAINT pk_roles PRIMARY KEY (role_id)
36 );
37
38
39 CREATE TABLE permissions (
40     permission_id    UUID                NOT NULL DEFAULT gen_random_uuid(),
41     permission_code   VARCHAR(50)        NOT NULL,
42     description      TEXT                NULL,
43     created_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
44     CONSTRAINT pk_permissions PRIMARY KEY (permission_id),
45     CONSTRAINT uq_permissions_code UNIQUE (permission_code)
46 );
47
48
49 CREATE TABLE github_connections (
50     connection_id    UUID                NOT NULL DEFAULT gen_random_uuid(),
51     user_id          UUID                NOT NULL,
52     github_user_id    VARCHAR(50)        NOT NULL,
53     github_username    VARCHAR(100)      NOT NULL,
54     access_token      VARCHAR(500)       NOT NULL,
55     created_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
56     updated_at       TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
57     CONSTRAINT pk_github_connections PRIMARY KEY (connection_id),
58     CONSTRAINT uq_github_connections_user UNIQUE (user_id),

```

```

59     CONSTRAINT fk_github_connections_user
60         FOREIGN KEY (user_id) REFERENCES users (user_id)
61         ON UPDATE CASCADE
62         ON DELETE CASCADE
63 );
64
65
66 CREATE TABLE user_notification_preferences (
67     preference_id    UUID                NOT NULL DEFAULT gen_random_uuid(),
68     user_id          UUID                NOT NULL,
69     channel          VARCHAR(20)         NOT NULL,
70     CONSTRAINT pk_user_notification_preferences PRIMARY KEY
71     (preference_id),
72     CONSTRAINT fk_user_notification_pref_user
73         FOREIGN KEY (user_id) REFERENCES users (user_id)
74         ON UPDATE CASCADE
75         ON DELETE CASCADE
76 );
77
78 CREATE TABLE github_connection_scopes (
79     scope_id         UUID                NOT NULL DEFAULT gen_random_uuid(),
80     connection_id    UUID                NOT NULL,
81     scope_value      VARCHAR(50)         NOT NULL,
82     CONSTRAINT pk_github_connection_scopes PRIMARY KEY (scope_id),
83     CONSTRAINT fk_github_connection_scopes_conn
84         FOREIGN KEY (connection_id) REFERENCES github_connections
85         (connection_id)
86         ON UPDATE CASCADE
87         ON DELETE CASCADE
88 );
89
90 CREATE TABLE workspace_members (
91     membership_id    UUID                NOT NULL DEFAULT gen_random_uuid(),
92     user_id          UUID                NOT NULL,
93     workspace_id     UUID                NOT NULL,
94     role_id          UUID                NOT NULL,
95     joined_at        TIMESTAMP           NOT NULL DEFAULT CURRENT_TIMESTAMP,
96     status           VARCHAR(20)         NOT NULL,
97     CONSTRAINT pk_workspace_members PRIMARY KEY (membership_id),
98     CONSTRAINT uq_workspace_members_user_workspace UNIQUE (user_id,
99     workspace_id),
100     CONSTRAINT fk_workspace_members_user
101         FOREIGN KEY (user_id) REFERENCES users (user_id)
102         ON UPDATE CASCADE
103         ON DELETE CASCADE,

```

```

103     CONSTRAINT fk_workspace_members_workspace
104         FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
105         ON UPDATE CASCADE
106         ON DELETE CASCADE,
107     CONSTRAINT fk_workspace_members_role
108         FOREIGN KEY (role_id) REFERENCES roles (role_id)
109         ON UPDATE CASCADE
110         ON DELETE RESTRICT
111 );
112
113
114 CREATE TABLE role_permissions (
115     role_permission_id UUID          NOT NULL DEFAULT gen_random_uuid(),
116     role_id             UUID          NOT NULL,
117     permission_id       UUID          NOT NULL,
118     granted_at          TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
119     CONSTRAINT pk_role_permissions PRIMARY KEY (role_permission_id),
120     CONSTRAINT uq_role_permissions_pair UNIQUE (role_id, permission_id),
121     CONSTRAINT fk_role_permissions_role
122         FOREIGN KEY (role_id) REFERENCES roles (role_id)
123         ON UPDATE CASCADE
124         ON DELETE CASCADE,
125     CONSTRAINT fk_role_permissions_permission
126         FOREIGN KEY (permission_id) REFERENCES permissions (permission_id)
127         ON UPDATE CASCADE
128         ON DELETE CASCADE
129 );
130
131
132 CREATE INDEX idx_github_connections_user_id ON github_connections(user_id);
133 CREATE INDEX idx_workspace_members_workspace_id ON
134     workspace_members(workspace_id);
135 CREATE INDEX idx_workspace_members_user_id ON workspace_members(user_id);
136 CREATE INDEX idx_role_permissions_role_id ON role_permissions(role_id);
137
138 CREATE OR REPLACE FUNCTION set_updated_at()
139 RETURNS TRIGGER AS $$
140 BEGIN
141     NEW.updated_at := CURRENT_TIMESTAMP;
142     RETURN NEW;
143 END;
144
145
146 CREATE TRIGGER trg_users_updated_at
147     BEFORE UPDATE ON users
148     FOR EACH ROW EXECUTE FUNCTION set_updated_at();

```

```
149
150 CREATE TRIGGER trg_workspaces_updated_at
151     BEFORE UPDATE ON workspaces
152     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
153
154 CREATE TRIGGER trg_roles_updated_at
155     BEFORE UPDATE ON roles
156     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
157
158 CREATE TRIGGER trg_github_connections_updated_at
159     BEFORE UPDATE ON github_connections
160     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
161
162 GO
```

chapter 2.

Phân hệ Workspace & Extension (Thành viên 2)

2.1. Phân hệ Workspace & Extension

2.1.1. Cú pháp Rút gọn (Logical Schema Syntax)

- workspaces (workspace_id UUID PK, name VARCHAR(150), description TEXT, status VARCHAR(20), created_at DATETIMEOFFSET, updated_at DATETIMEOFFSET, deleted_at DATETIMEOFFSET)
- github_connections (connection_id UUID PK, workspace_id UUID FK, provider VARCHAR(50), account_name VARCHAR(100), access_token TEXT, sync_status VARCHAR(20), created_at DATETIME2, updated_at DATETIME2)
- github_connection_scopes (scope_id UUID PK, connection_id UUID FK, scope_value VARCHAR(100), created_at DATETIME2)
- repositories (repository_id UUID PK, workspace_id UUID FK, name VARCHAR(150), url VARCHAR(255), default_branch VARCHAR(50), created_at DATETIME2, updated_at DATETIME2)
- branches (branch_id UUID PK, repository_id UUID FK, name VARCHAR(100), is_protected BOOLEAN, created_at DATETIME2)

2.1.2. Mã DDL SQL Hoàn Chính (schema_member_workspace.sql)

```
1  -- 1. BANG WORKSPACES (Quan ly khong gian lam viec cua nguoi dung)
2  CREATE TABLE workspaces (
3      workspace_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
4      name VARCHAR(150) NOT NULL,
5      description TEXT NULL,
6      status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
7      created_at DATETIMEOFFSET DEFAULT SYSDATETIMEOFFSET(),
8      updated_at DATETIMEOFFSET DEFAULT SYSDATETIMEOFFSET(),
9      deleted_at DATETIMEOFFSET NULL,
10
11     -- Constraints
12     CONSTRAINT uk_workspace_id_name UNIQUE (workspace_id, name),
13     CONSTRAINT chk_workspace_status CHECK (status IN ('ACTIVE', 'ARCHIVED'))
14 );
15
16 -- 2. BANG GITHUB_CONNECTIONS (Ket noi tai khoan GitHub voi Workspace)
17 CREATE TABLE github_connections (
18     connection_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
19     workspace_id UNIQUEIDENTIFIER NOT NULL,
20     provider VARCHAR(50) NOT NULL DEFAULT 'GITHUB',
21     account_name VARCHAR(100) NOT NULL,
22     access_token TEXT NOT NULL,
23     sync_status VARCHAR(20) NOT NULL DEFAULT 'IDLE',
24     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
25     updated_at DATETIME2 NULL,
26
27     -- Constraints
28     CONSTRAINT uk_github_connection UNIQUE (connection_id, provider),
29     CONSTRAINT chk_sync_status CHECK (sync_status IN ('IDLE', 'SYNCING', 'SUCCESS', 'FAILED'))
30 );
31
32 -- 3. BANG GITHUB_CONNECTION_SCOPES (Pham vi quyen han cua ket noi GitHub)
33 CREATE TABLE github_connection_scopes (
34     scope_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
35     connection_id UNIQUEIDENTIFIER NOT NULL,
36     scope_value VARCHAR(100) NOT NULL,
37     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
38
39     CONSTRAINT uk_connection_scope UNIQUE (connection_id, scope_value)
40 );
41
42 -- 4. BANG REPOSITORIES (Kho luu tru ma nguon lien ket)
43 CREATE TABLE repositories (
```

```

44     repository_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
45     workspace_id UNIQUEIDENTIFIER NOT NULL,
46     name VARCHAR(150) NOT NULL,
47     url VARCHAR(255) NOT NULL,
48     default_branch VARCHAR(50) DEFAULT 'main',
49     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
50     updated_at DATETIME2 NULL,
51
52     CONSTRAINT uk_repository_name UNIQUE (repository_id, name)
53 );
54
55 -- 5. BANG BRANCHES (Cac nhanh lam viec trong Repository)
56 CREATE TABLE branches (
57     branch_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
58     repository_id UNIQUEIDENTIFIER NOT NULL,
59     name VARCHAR(100) NOT NULL,
60     is_protected BIT NOT NULL DEFAULT 0,
61     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
62
63     CONSTRAINT uk_branch_name UNIQUE (repository_id, name)
64 );
65 GO
66
67 -- =====
68 -- KHAI BAO CAC KHOA NGOAI THAM CHIEU (CROSS-DOMAIN & INTERNAL FKs)
69 -- =====
70 ALTER TABLE github_connections ADD CONSTRAINT
71     fk_github_connections_workspace
72     FOREIGN KEY (workspace_id) REFERENCES workspaces(workspace_id) ON
73     DELETE CASCADE;
74
75 ALTER TABLE github_connection_scopes ADD CONSTRAINT
76     fk_github_scopes_connection
77     FOREIGN KEY (connection_id) REFERENCES github_connections(connection_id)
78     ON DELETE CASCADE;
79
80 ALTER TABLE repositories ADD CONSTRAINT fk_repositories_workspace
81     FOREIGN KEY (workspace_id) REFERENCES workspaces(workspace_id) ON
82     DELETE CASCADE;
83
84 ALTER TABLE branches ADD CONSTRAINT fk_branches_repository
85     FOREIGN KEY (repository_id) REFERENCES repositories(repository_id) ON
86     DELETE CASCADE;
87
88 GO

```

2.2. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

2.2.1. Cú pháp Rút gọn (Logical Schema Syntax)

- `templates` (template_id UUID PK, workspace_id UUID FK, name VARCHAR(150), description TEXT, content_structure TEXT, creator_type VARCHAR(20), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP)
- `template_sections` (section_id UUID PK, template_id UUID FK, heading_title VARCHAR(255), order_index INTEGER, prompt_instruction TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- `placeholders` (placeholder_id UUID PK, section_id UUID FK, variable_name VARCHAR(100), expected_data_type VARCHAR(30), description TEXT, is_required BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- `template_mappings` (mapping_id UUID PK, placeholder_id UUID FK, code_entity_id UUID FK, extraction_rule TEXT, is_active BOOLEAN)
- `documents` (document_id UUID PK, workspace_id UUID FK, author_user_id UUID FK, title VARCHAR(255), document_type VARCHAR(30), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)
- `document_versions` (version_id UUID PK, document_id UUID FK, version_number VARCHAR(20), content TEXT, change_summary TEXT, status VARCHAR(20), ticket_codes VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)
- `change_logs` (log_id UUID PK, version_id UUID FK, change_type VARCHAR(20), change_description TEXT, diff_content TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)

2.2.2. Mã DDL SQL Hoàn Chỉnh (schema_member_4.sql)

```
1
2
3
4 CREATE TABLE templates (
5     template_id      UUID          NOT NULL DEFAULT gen_random_uuid(),
6     workspace_id     UUID          NOT NULL,
7     name             VARCHAR(150)  NOT NULL,
8     description      TEXT          NULL,
9     content_structure TEXT          NULL,
10    creator_type      VARCHAR(20)   NOT NULL,
11    status            VARCHAR(20)   NOT NULL DEFAULT 'DRAFT',
```

```

12     created_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
13     updated_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
14     CONSTRAINT pk_templates PRIMARY KEY (template_id),
15
16     CONSTRAINT uq_templates_workspace_name UNIQUE (workspace_id, name),
17     CONSTRAINT fk_templates_workspace
18         FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
19         ON UPDATE CASCADE
20         ON DELETE CASCADE,
21     CONSTRAINT ck_templates_creator_type
22         CHECK (creator_type IN ('SYSTEM', 'USER', 'AI')),
23     CONSTRAINT ck_templates_status
24         CHECK (status IN ('DRAFT', 'ACTIVE', 'ARCHIVED')),
25     CONSTRAINT ck_templates_name_not_blank
26         CHECK (length(btrim(name)) > 0),
27     CONSTRAINT ck_templates_updated_after_created
28         CHECK (updated_at >= created_at)
29 );
30
31 CREATE TABLE template_sections (
32     section_id          UUID              NOT NULL DEFAULT gen_random_uuid(),
33     template_id         UUID              NOT NULL,
34     heading_title       VARCHAR(255)      NOT NULL,
35     order_index         INTEGER           NOT NULL,
36     prompt_instruction  TEXT              NULL,
37     created_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
38     updated_at          TIMESTAMP          NOT NULL DEFAULT CURRENT_TIMESTAMP,
39     CONSTRAINT pk_template_sections PRIMARY KEY (section_id),
40
41     CONSTRAINT uq_template_sections_order UNIQUE (template_id,
42     order_index),
43     CONSTRAINT fk_template_sections_template
44         FOREIGN KEY (template_id) REFERENCES templates (template_id)
45         ON UPDATE CASCADE
46         ON DELETE CASCADE,
47     CONSTRAINT ck_template_sections_order_index
48         CHECK (order_index >= 0),
49     CONSTRAINT ck_template_sections_heading_not_blank
50         CHECK (length(btrim(heading_title)) > 0)
51 );
52
53 CREATE TABLE placeholders (
54     placeholder_id      UUID              NOT NULL DEFAULT gen_random_uuid(),
55     section_id          UUID              NOT NULL,
56     variable_name       VARCHAR(100)      NOT NULL,
57     expected_data_type  VARCHAR(30)       NOT NULL,
58     description         TEXT              NULL,

```

```

58     is_required          BOOLEAN          NOT NULL DEFAULT TRUE,
59     created_at           TIMESTAMP        NOT NULL DEFAULT CURRENT_TIMESTAMP,
60     updated_at           TIMESTAMP        NOT NULL DEFAULT CURRENT_TIMESTAMP,
61     CONSTRAINT pk_placeholders PRIMARY KEY (placeholder_id),
62
63     CONSTRAINT uq_placeholders_section_variable UNIQUE (section_id,
64     variable_name),
65     CONSTRAINT fk_placeholders_section
66         FOREIGN KEY (section_id) REFERENCES template_sections (section_id)
67         ON UPDATE CASCADE
68         ON DELETE CASCADE,
69     CONSTRAINT ck_placeholders_expected_data_type
70         CHECK (expected_data_type IN ('STRING', 'NUMBER', 'BOOLEAN',
71         'DATE', 'LIST', 'OBJECT', 'CODE')),
72
73     CONSTRAINT ck_placeholders_variable_name_format
74         CHECK (variable_name ~ '^[A-Za-z_][A-Za-z0-9_]*$')
75 );
76
77 CREATE TABLE template_mappings (
78     mapping_id           UUID             NOT NULL DEFAULT gen_random_uuid(),
79     placeholder_id       UUID             NOT NULL,
80     code_entity_id       UUID             NOT NULL,
81     extraction_rule      TEXT             NULL,
82     is_active            BOOLEAN          NOT NULL DEFAULT TRUE,
83     CONSTRAINT pk_template_mappings PRIMARY KEY (mapping_id),
84
85     CONSTRAINT uq_template_mappings_pair UNIQUE (placeholder_id,
86     code_entity_id),
87     CONSTRAINT fk_template_mappings_placeholder
88         FOREIGN KEY (placeholder_id) REFERENCES placeholders
89         (placeholder_id)
90         ON UPDATE CASCADE
91         ON DELETE CASCADE,
92     CONSTRAINT fk_template_mappings_code_entity
93         FOREIGN KEY (code_entity_id) REFERENCES code_entities
94         (code_entity_id)
95         ON UPDATE CASCADE
96         ON DELETE CASCADE
97 );
98
99 CREATE TABLE documents (
100     document_id          UUID             NOT NULL DEFAULT gen_random_uuid(),
101     workspace_id          UUID             NOT NULL,
102     author_user_id        UUID             NOT NULL,
103     title                 VARCHAR(255)     NOT NULL,
104     document_type          VARCHAR(30)     NOT NULL,

```

```

101     status          VARCHAR(20)          NOT NULL DEFAULT 'DRAFT',
102     created_at      TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
103     updated_at      TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
104     deleted_at      TIMESTAMP            NULL,
105     CONSTRAINT pk_documents PRIMARY KEY (document_id),
106     CONSTRAINT fk_documents_workspace
107         FOREIGN KEY (workspace_id) REFERENCES workspaces (workspace_id)
108         ON UPDATE CASCADE
109         ON DELETE CASCADE,
110     CONSTRAINT fk_documents_author
111         FOREIGN KEY (author_user_id) REFERENCES users (user_id)
112         ON UPDATE CASCADE
113         ON DELETE RESTRICT,
114     CONSTRAINT ck_documents_type
115         CHECK (document_type IN ('SRS', 'API_DOC', 'USER_GUIDE',
116                                   'RELEASE_NOTE', 'TECH_DESIGN', 'OTHER')),
117     CONSTRAINT ck_documents_status
118         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'PUBLISHED', 'ARCHIVED')),
119     CONSTRAINT ck_documents_title_not_blank
120         CHECK (length(btrim(title)) > 0),
121     CONSTRAINT ck_documents_deleted_after_created
122         CHECK (deleted_at IS NULL OR deleted_at >= created_at)
123 );
124
125 CREATE UNIQUE INDEX uq_documents_workspace_title_active
126     ON documents (workspace_id, title)
127     WHERE deleted_at IS NULL;
128
129 CREATE TABLE document_versions (
130     version_id      UUID                NOT NULL DEFAULT gen_random_uuid(),
131     document_id     UUID                NOT NULL,
132     version_number   VARCHAR(20)        NOT NULL,
133     content         TEXT                NULL,
134     change_summary   TEXT                NULL,
135     status          VARCHAR(20)        NOT NULL DEFAULT 'DRAFT',
136     ticket_codes     VARCHAR(500)       NULL,
137     created_at      TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
138     updated_at      TIMESTAMP            NOT NULL DEFAULT CURRENT_TIMESTAMP,
139     CONSTRAINT pk_document_versions PRIMARY KEY (version_id),
140
141     CONSTRAINT uq_document_versions_number UNIQUE (document_id,
142     version_number),
143     CONSTRAINT fk_document_versions_document
144         FOREIGN KEY (document_id) REFERENCES documents (document_id)
145         ON UPDATE CASCADE
146         ON DELETE CASCADE,
147     CONSTRAINT ck_document_versions_status

```

```

147         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'APPROVED', 'REJECTED',
148         'PUBLISHED')),
149
150     CONSTRAINT ck_document_versions_number_format
151         CHECK (version_number ~ '^[vV]?[0-9]+(\.[0-9]+){0,2}$')
152 );
153
154 CREATE TABLE change_logs (
155     log_id            UUID            NOT NULL DEFAULT gen_random_uuid(),
156     version_id        UUID            NOT NULL,
157     change_type        VARCHAR(20)    NOT NULL,
158     change_description TEXT            NULL,
159     diff_content       TEXT            NULL,
160     created_at         TIMESTAMP       NOT NULL DEFAULT CURRENT_TIMESTAMP,
161     updated_at         TIMESTAMP       NOT NULL DEFAULT CURRENT_TIMESTAMP,
162     CONSTRAINT pk_change_logs PRIMARY KEY (log_id),
163     CONSTRAINT fk_change_logs_version
164         FOREIGN KEY (version_id) REFERENCES document_versions (version_id)
165         ON UPDATE CASCADE
166         ON DELETE CASCADE,
167     CONSTRAINT ck_change_logs_change_type
168         CHECK (change_type IN ('CREATE', 'UPDATE', 'DELETE', 'RESTORE',
169         'PUBLISH'))
170 );
171
172 CREATE INDEX idx_templates_workspace_id ON templates (workspace_id);
173
174 CREATE INDEX idx_templates_status ON templates (status);
175
176 CREATE INDEX idx_template_mappings_code_entity ON template_mappings
177     (code_entity_id);
178
179 CREATE INDEX idx_template_mappings_active ON template_mappings
180     (placeholder_id)
181     WHERE is_active = TRUE;
182
183 CREATE INDEX idx_documents_workspace_id ON documents (workspace_id);
184
185 CREATE INDEX idx_documents_author ON documents (author_user_id);
186
187 CREATE INDEX idx_documents_status ON documents (status);
188
189 CREATE INDEX idx_documents_created_at ON documents (created_at DESC);
190
191 CREATE INDEX idx_document_versions_status ON document_versions
192     (status);
193
194 CREATE INDEX idx_document_versions_created_at ON document_versions
195     (document_id, created_at DESC);
196
197 CREATE INDEX idx_change_logs_version_id ON change_logs (version_id);
198
199 CREATE INDEX idx_change_logs_created_at ON change_logs (created_at DESC);

```

```

188
189 CREATE OR REPLACE FUNCTION set_updated_at()
190 RETURNS TRIGGER AS $$
191 BEGIN
192     NEW.updated_at := CURRENT_TIMESTAMP;
193     RETURN NEW;
194 END;
195
196 CREATE TRIGGER trg_templates_updated_at
197     BEFORE UPDATE ON templates
198     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
199 CREATE TRIGGER trg_template_sections_updated_at
200     BEFORE UPDATE ON template_sections
201     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
202 CREATE TRIGGER trg_placeholders_updated_at
203     BEFORE UPDATE ON placeholders
204     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
205 CREATE TRIGGER trg_documents_updated_at
206     BEFORE UPDATE ON documents
207     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
208 CREATE TRIGGER trg_document_versions_updated_at
209     BEFORE UPDATE ON document_versions
210     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
211 CREATE TRIGGER trg_change_logs_updated_at
212     BEFORE UPDATE ON change_logs
213     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
214 GO

```

2.3. Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

2.3.1. Cú pháp Rút gọn (Logical Schema Syntax)

- reviews (review_id UUID PK, version_id UUID FK, user_id UUID FK, status VARCHAR(20), decision VARCHAR(20), summary_notes TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- review_comments (comment_id UUID PK, review_id UUID FK, user_id UUID FK, content TEXT, is_resolved BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- approvals (approval_id UUID PK, version_id UUID FK, user_id UUID FK, decision VARCHAR(20), rejection_reason TEXT, approved_at TIMESTAMP)
- grounding_evidences (evidence_id UUID PK, version_id UUID FK, snippet_content TEXT, purpose VARCHAR(30), created_at TIMESTAMP)

- drift_alerts (alert_id UUID PK, evidence_id UUID FK, repository_id UUID FK, drift_type VARCHAR(20), status VARCHAR(20), detected_at TIMESTAMP)

2.3.2. Mã DDL SQL Hoàn Chính (schema_member_5.sql)

```

1
2
3 CREATE TABLE reviews (
4     review_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
5     version_id UNIQUEIDENTIFIER NOT NULL,
6     user_id UNIQUEIDENTIFIER NOT NULL,
7     status VARCHAR(20) NOT NULL DEFAULT 'PENDING',
8     decision VARCHAR(20) NULL,
9     summary_notes NVARCHAR(MAX) NULL,
10    created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
11    updated_at DATETIME2 NULL,
12
13    CONSTRAINT chk_reviews_status CHECK (status IN ('PENDING', 'IN_PROGRESS', 'COMPLETED')),
14    CONSTRAINT chk_reviews_decision CHECK (decision IS NULL OR decision IN ('APPROVED', 'REJECTED', 'CHANGES_REQUESTED'))
15 );
16
17 CREATE TABLE review_comments (
18     comment_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
19     review_id UNIQUEIDENTIFIER NOT NULL,
20     user_id UNIQUEIDENTIFIER NOT NULL,
21     content NVARCHAR(MAX) NOT NULL,
22     is_resolved BIT NOT NULL DEFAULT 0,
23     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
24     updated_at DATETIME2 NULL,
25
26     CONSTRAINT fk_comments_review FOREIGN KEY (review_id)
27         REFERENCES reviews(review_id) ON DELETE CASCADE
28 );
29
30 CREATE TABLE approvals (
31     approval_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
32     version_id UNIQUEIDENTIFIER NOT NULL,
33     user_id UNIQUEIDENTIFIER NOT NULL,
34     decision VARCHAR(20) NOT NULL,
35     rejection_reason NVARCHAR(MAX) NULL,
36     approved_at DATETIME2 NOT NULL DEFAULT GETDATE(),
37
38     CONSTRAINT chk_approvals_decision CHECK (decision IN ('APPROVED', 'REJECTED'))
39 );

```

```

40
41 CREATE TABLE grounding_evidences (
42     evidence_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
43     version_id UNIQUEIDENTIFIER NOT NULL,
44     snippet_content NVARCHAR(MAX) NOT NULL,
45     purpose VARCHAR(30) NOT NULL DEFAULT 'HALLUCINATION_PREVENTION',
46     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
47
48     CONSTRAINT chk_evidences_purpose CHECK (purpose IN ('
HALLUCINATION_PREVENTION', 'CODE_REFERENCE', 'AST_VERIFICATION'))
49 );
50
51 CREATE TABLE drift_alerts (
52     alert_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
53     evidence_id UNIQUEIDENTIFIER NOT NULL,
54     repository_id UNIQUEIDENTIFIER NOT NULL,
55     drift_type VARCHAR(20) NOT NULL,
56     status VARCHAR(20) NOT NULL DEFAULT 'OPEN',
57     detected_at DATETIME2 NOT NULL DEFAULT GETDATE(),
58
59     CONSTRAINT fk_alerts_evidence FOREIGN KEY (evidence_id)
60         REFERENCES grounding_evidences(evidence_id) ON DELETE CASCADE,
61     CONSTRAINT chk_alerts_type CHECK (drift_type IN ('REFERENTIAL', '
SIGNATURE', 'SEMANTIC', 'OUTDATED')),
62     CONSTRAINT chk_alerts_status CHECK (status IN ('OPEN', 'RESOLVED', '
IGNORED'))
63 );
64
65
66
67 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_version
68     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
69 ALTER TABLE reviews ADD CONSTRAINT fk_reviews_user
70     FOREIGN KEY (user_id) REFERENCES users(user_id);
71 ALTER TABLE review_comments ADD CONSTRAINT fk_comments_user
72     FOREIGN KEY (user_id) REFERENCES users(user_id);
73 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_version
74     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
75 ALTER TABLE approvals ADD CONSTRAINT fk_approvals_user
76     FOREIGN KEY (user_id) REFERENCES users(user_id);
77 ALTER TABLE grounding_evidences ADD CONSTRAINT fk_evidences_version
78     FOREIGN KEY (version_id) REFERENCES document_versions(version_id);
79 ALTER TABLE drift_alerts ADD CONSTRAINT fk_alerts_repository
80     FOREIGN KEY (repository_id) REFERENCES repositories(repository_id);
81
82 GO

```


chapter 3.

Future research